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Data-efficient pathology whole-slide image segmentation with slide-level conditioning

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Lunchi Guo [1], Dominique Boucher[2,3], Mireille I. Ouimet [2,3] & Christopher Sun[1,3]

Whole-slide images (WSIs) are inherently high-resolution, gigapixel-scale data that require models to reconcile global slide context with fine-grained patch precision, yet progress is often limited by scarce pixel-level annotations. In this work, we address binary foreground segmentation of two histopathological target structures: glomeruli in PAS-stained kidney WSIs and atherosclerotic plaque lesions in murine aortic root sections, where the foreground corresponds to the dataset-specific structure of interest and all remaining tissue and background are treated as non-target regions. We introduce SlideFiLM, a segmentation framework that uses Prov-GigaPath whole-slide representations to guide tile decoding with slide context through a bounded slide-conditioned modulation operator that is shared across tiles. By injecting slide-level context across all tiles, the method achieves end-to-end segmentation on gigapixel WSIs. Across both benchmarks, SlideFiLM outperforms established baselines, with the largest gains for tissue structures that require global tissue context. Crucially, SlideFiLM maintains accuracy in low-data regimes, achieving robust performance under reduced training sets, and shows enhanced slide-level consistency as well as robustness across tissue types. These findings highlight SlideFiLM as an effective solution for WSI segmentation that incorporates global context, providing complete slide coverage, stable performance with limited labels, and consistent gains on structures that depend on broader contextual information.

Digitization of histopathology slides into whole-slide images (WSIs) has enabled computational pathology at gigapixel resolution, supporting diagnosis, classification, and biomarker discovery[1]. Clinically and biologically, segmentation (e.g., delineation of tissue compartments and structures) enables burden estimation, morphometric biomarkers, and the generation of labels for downstream slide-level analysis. However, producing pixel-accurate segmentation across WSIs remains labour-intensive, requiring domain expertise and substantial annotation time, and therefore limits the scale and reproducibility of segmentation-based pipelines.

Advances in deep learning have established the effectiveness of fully convolutional architectures for biomedical segmentation, setting the groundwork for current WSI segmentation practice[2–4]. However, whole-slide images are typically very large, often containing millions to billions of pixels, which makes direct end-to-end segmentation computationally expensive and memory intensive. Indeed, model training and inference are typically performed on smaller tiles that are later stitched together to meet computational memory constraints[5]. Tile-centric pipelines remain widely used for their efficiency and scalability. Yet, their narrow field of view prevents access to global slide semantics that are essential for context-aware delineation. This purely local perspective often leads to fragmented predictions and boundary inconsistencies across tiles, emphasizing the need for segmentation frameworks that integrate global contextual information while retaining patch-level efficiency. To expand the effective context, recent approaches enhance local decoding by incorporating information at multiple scales or across regions[6–10]. Instead of relying solely on isolated tiles, they introduce hierarchical or dual branch designs that exchange features between detailed and contextual views, enabling the model to capture structural details as well as global semantics. Such strategies markedly improve boundary coherence and object delineation while maintaining computational efficiency. However, most existing local-global fusion methods still derive contextual information from parallel spatial image branches, leaving the global context constrained by the spatial regions sampled during tile processing rather than by an independent slide-level representation.

Despite recent advances, a central practical constraint is the scarcity and high cost of expert annotations at the pixel level, and models trained

[1]Telfer School of Management, University of Ottawa, Ottawa, ON, Canada. [2]Department of Biochemistry, Microbiology and Immunology, University of Ottawa, Ottawa, ON, Canada. [3]University of Ottawa Heart Institute, University of Ottawa, Ottawa, ON, Canada. e-mail: sun@telfer.uottawa.ca

with limited supervision often struggle to generalize across diverse tissue types and staining variations. Existing label-efficient, self-supervised, and weakly-supervised approaches alleviate annotation demands by transferring knowledge from large-scale pretraining or by learning intrinsic visual correspondences within WSIs[11–16]. These methods improve data efficiency and generalization, yet their generalization remains largely at the feature level, as they learn local patch correspondences without explicit mechanisms to align slide-level semantics or contextual dependencies. Recent prompt-based segmentation models, such as Segment Anything Model (SAM)[17] and its microscopy-oriented variants including μSAM[18], illustrate that foundation pretraining can reduce annotation requirements and enable transfer to new imaging conditions. However, when applied to histopathology, performance often degrades without domain adaptation, likely due to distribution shift in tissue texture, stain variability, and contrast, as well as the multi-scale context required for whole-slide analysis. In contrast, Prov-GigaPath is the first WSI foundation model with open weights, trained on more than one billion tiles from real-world slides. It shows that large scale self-supervised pretraining, together with long context modeling, can substantially improve downstream pathology tasks across tissues and institutions[19]. Although Prov-GigaPath offers strong generalization across tissues and institutions, its design is optimized for slide-level inference rather than spatially resolved prediction. The absence of pixel-level supervision and slide-to-tile alignment constrains its utility for precise tissue delineation.

Here we introduce SlideFiLM, a slide-conditioned framework for whole-slide segmentation that addresses the dual challenge of reconciling slide-wide tissue context with tile-level precision while remaining effective when dense pixel annotations are limited. Unlike conventional local-global fusion methods that combine image features across magnifications or spatial branches, SlideFiLM conditions tile-level decoding on a compact whole-slide representation derived from a pretrained WSI foundation model. This representation is injected into the decoder through feature-wise linear modulation (FiLM), enabling dense predictions that remain consistent with the broader tissue architecture of the slide. To support learning in low-data regimes, we implement this slide-to-tile conditioning paradigm using representations from a pretrained whole-slide foundation model (Prov-GigaPath), adapting it for dense segmentation tasks. Concretely, we formulate the task as binary foreground segmentation of two histopathological target structures: glomeruli in PAS-stained kidney WSIs from the Kidney Pathology Image segmentation (KPIs) dataset[20], and atherosclerotic plaque lesions in murine aortic root sections from our newly curated Mouse Aortic Root Atherosclerosis (MARA) dataset, where the foreground corresponds to the dataset-specific structure of interest and all remaining tissue and background are treated as non-target regions. These two datasets provide complementary test settings for slide-level conditioning. MARA represents a context-sensitive task in which plaque identification is linked to broader aortic-root anatomy, whereas KPIs serves as an established public segmentation benchmark in which glomeruli can often be identified from local morphology, allowing us to assess whether slide-level conditioning also provides benefits when local visual cues are already informative. We validate SlideFiLM on these datasets, demonstrating robust performance under low-data training and across distinct tissue types and staining conditions. The approach provides complete slide coverage with pixel-aligned outputs, achieves competitive performance against strong baselines, and maintains accuracy when trained on reduced training sets. These findings support pairing pretrained slide embeddings with explicit slide-to-tile conditioning as a simple and deployment-friendly path to accurate and label-efficient WSI segmentation.

Results

SlideFiLM overview

SlideFiLM is a slide-conditioned segmentation framework for WSIs, tailored to low-data regimes with limited dense pixel labels, where purely tile-local decoders often struggle to disambiguate morphologies that require global context. Notably, WSI segmentation is often constrained by the mismatch between whole slide context and the limited field of view of tile based encoders, as conventional models process tiles in isolation and therefore cannot reliably leverage slide context that helps disambiguate tissue identity and morphology. As illustrated in Fig. 1a, a Vision Transformer (ViT) tile encoder maps each tile to (i) patch-token features that retain spatial layout for dense reconstruction and (ii) a compact tile embedding for slide-level aggregation; a long-context slide encoder then integrates the sequence of tile embeddings into a global slide embedding summarizing tissue composition and slide-wide morphology. The central methodological contribution is a bidirectional coupling between these representations: the slide embedding is transformed into FiLM conditioning parameters (γ, β) [21] and injected at multiple stages of an upsampling decoder (Fig. 1b), so that per-tile dense features are modulated by whole-slide context while being decoded into pixel-aligned logits. Multi-level skip connections from intermediate encoder blocks preserve high-frequency detail for boundary refinement, while reusing the same slide-level conditioning across all tiles promotes slide-consistent predictions and improved label efficiency. In our implementation, the tile-slide backbone is instantiated with Prov-GigaPath[19], but the formulation is model-agnostic and applies to any WSI representation model that provides tile embeddings and patch-level features.

SlideFiLM improves accuracy and completeness

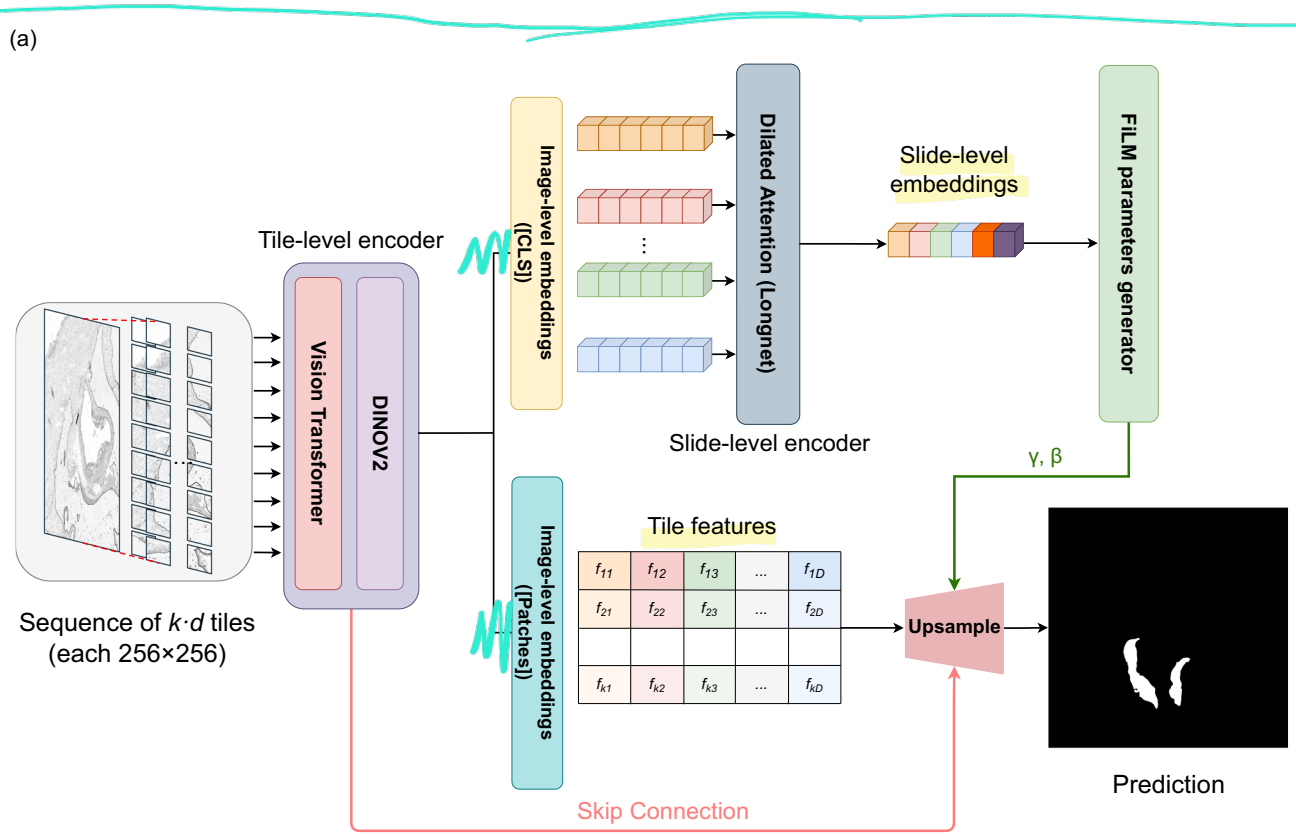
To evaluate SlideFiLM's overall segmentation capability, we benchmarked it against established baseline models on the KPIs dataset and a newly annotated real-world histology dataset from the Ottawa Heart Institute, the MARA segmentation dataset. All models were trained using identical data splits and evaluated on test sets, with Dice reported as the primary metric. SlideFiLM consistently outperformed baseline models, including U-Net, DeepLabv3+[22], SwinUNETR[23], SegFormer[24], SAM, and HookNet. These methods were chosen to cover convolutional (U-Net, DeepLabv3+), transformer-based (SegFormer, SwinUNETR), foundation-model-based (SAM), and multi-resolution histopathology segmentation (HookNet) paradigms. For the evaluation of the dataset, we calculated metrics in each patch. In addition, to show the effectiveness of the WSI-level segmentation, we computed the Dice score on the entire WSI foreground. The Dice scores across each cohort were also averaged to calculate the overall patch-level and WSI-level Dice scores, providing a holistic measure of performance for each dataset.

On the KPIs dataset (Table 1), SlideFiLM achieved the best overall performance, with a patch-level Dice of 92.96 ± 0.73 and a WSI-level Dice of 90.03 ± 0.71. On the MARA dataset (Table 2), SlideFiLM again obtained the best scores, reaching a patch-level Dice of 89.77 ± 0.43 and a WSI-level Dice of 91.78 ± 0.47. For each method, experiments were repeated with three different random seeds, and the results are reported as mean ± standard deviation across the three runs. Together, these evaluations suggest that SlideFiLM performs promisingly across two murine histopathology datasets differing in tissue type, segmentation target, staining protocol, and scanner.

Qualitative results (Fig. 2) further demonstrate that SlideFiLM produces smoother and more complete segmentation maps. Compared with baseline methods, SlideFiLM better preserves thin structures and avoids spurious holes or discontinuities within large tissue regions. The integration of multi-layer skip connections supplies fine-scale texture cues during upsampling, while FiLM-based slide conditioning enforces consistency across tiles, yielding morphologically coherent predictions.

WSI-level boundary and structural consistency evaluation

To substantiate the qualitative observation that SlideFiLM produces more complete structures and fewer spurious fragments, we further evaluated WSI-level boundary accuracy and structural consistency using structure-aware metrics. In addition to WSI-level Dice, we computed Boundary F1 and 95% Hausdorff Distance (HD95) for both datasets. For the KPIs dataset, which contains many spatially separated glomerular regions, we additionally computed object-level F1 based on connected-component matching to assess whether the model preserves individual positive structures and reduces fragmented or spurious predictions.



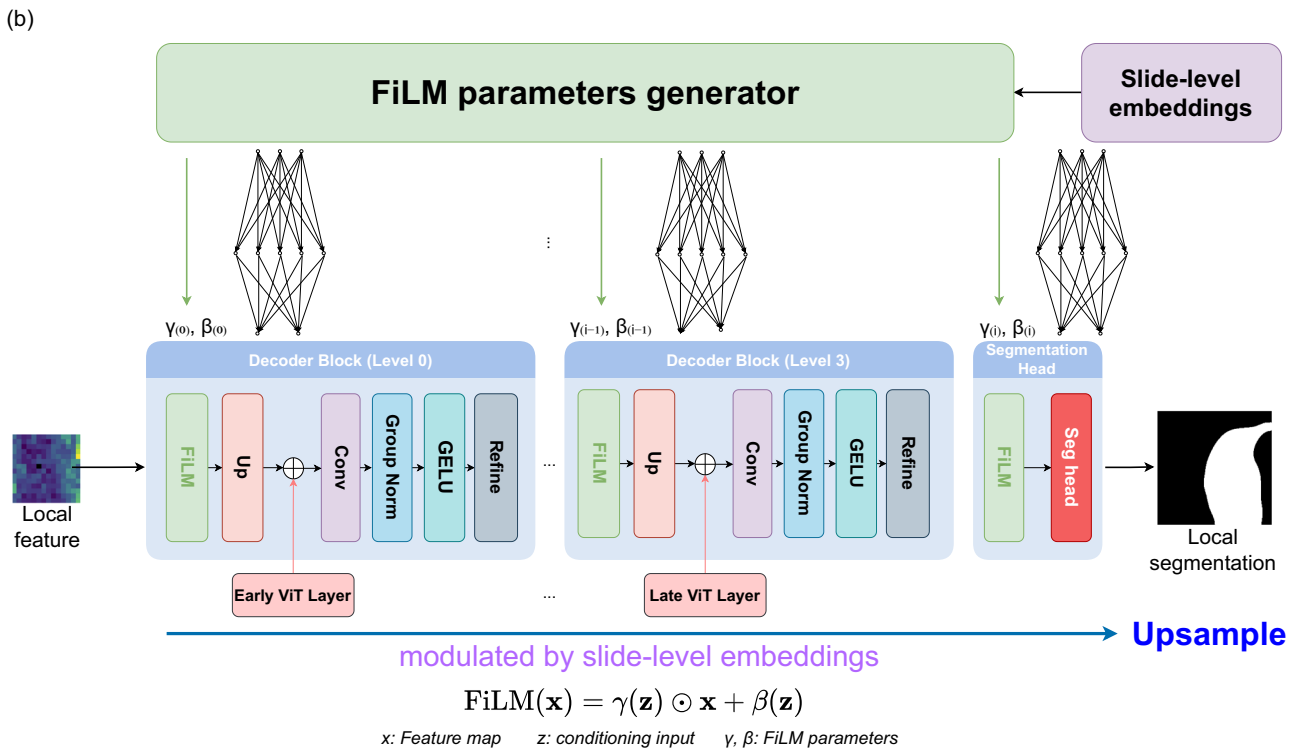


Fig. 1 | SlideFiLM architecture and upsampling mechanism. a SlideFiLM model architecture. Each 256 × 256 tile from the whole-slide image is encoded using a DINOv2 Vision Transformer to extract both patch-level and [CLS]-level embeddings. The [CLS] embeddings are aggregated by a dilated-attention LongNet slide-level encoder to form a global slide embedding, which is used to generate FiLM modulation parameters (γ, β). Patch-level embeddings are decoded through an upsampling pathway conditioned by these FiLM parameters, with skip connections from tile-level features to preserve spatial detail. **b** Illustration of the FiLM-modulated upsampling decoder. Each decoder block applies FiLM modulation using slide-level embeddings and fuses features from corresponding ViT layers. Early ViT features supply spatial details, while deeper layers provide semantic context, enabling hierarchical and context-aware segmentation refinement.

Table 1 | Cross-cohort segmentation performance on the KPIs dataset

Method	Patch-level Dice by cohort				Average	
	5/6Nx	DN	NEP25	Normal	Patch-level Dice	WSI-level Dice
U-Net	85.94 ± 0.63	87.81 ± 0.78	87.24 ± 0.84	89.02 ± 0.57	87.50 ± 0.69	80.18 ± 0.91
DeepLabv3+	87.10 ± 0.58	89.58 ± 0.74	87.52 ± 0.66	89.36 ± 0.81	88.39 ± 0.62	81.27 ± 0.88
SegFormer	90.11 ± 0.49	90.88 ± 0.67	91.94 ± 0.73	89.84 ± 0.56	90.69 ± 0.61	85.43 ± 0.79
SwinUNETR-V2	89.59 ± 0.71	90.84 ± 0.83	89.27 ± 0.92	89.51 ± 0.64	89.80 ± 0.76	85.57 ± 0.56
SAM-ViT-B	89.45 ± 0.68	89.66 ± 0.59	89.61 ± 0.77	89.73 ± 0.52	89.61 ± 0.65	84.31 ± 0.89
SAM-ViT-H	90.68 ± 0.74	90.51 ± 0.63	89.67 ± 0.81	89.53 ± 0.69	90.10 ± 0.58	84.82 ± 0.93
HookNet	91.17 ± 0.47	92.03 ± 0.64	89.77 ± 0.81	91.23 ± 0.58	91.05 ± 0.65	87.49 ± 0.94
SlideFiLM	**94.20 ± 0.85**	**92.10 ± 0.68**	**92.45 ± 0.82**	**93.30 ± 0.74**	**92.96 ± 0.73**	**90.03 ± 0.71**

Cross-cohort segmentation performance on KPIs. We evaluate glomerulus segmentation across four clinically distinct kidney pathology cohorts. Dice similarity coefficient scores (%) are reported. Patch Dice represents direct evaluation on high-resolution non-overlapping patches obtained from each slide.

Table 2 | Cross-cohort segmentation performance on the MARA dataset

Method	Patch-level Dice by cohort		Average	
	Oil Red O	Masson's Trichrome	Patch-level Dice	WSI-level Dice
U-Net	84.12 ± 0.68	86.39 ± 0.84	85.26 ± 0.57	83.31 ± 0.93
DeepLabv3+	85.31 ± 0.76	86.43 ± 0.61	85.87 ± 0.88	84.91 ± 1.04
SegFormer	87.46 ± 0.52	88.29 ± 0.73	87.88 ± 0.66	87.12 ± 0.81
SwinUNETR-V2	87.78 ± 0.89	89.03 ± 0.74	88.41 ± 1.02	87.49 ± 0.67
SAM-ViT-B	86.68 ± 0.71	87.78 ± 0.59	87.23 ± 0.86	84.63 ± 1.08
SAM-ViT-H	85.72 ± 0.82	87.37 ± 0.64	86.55 ± 0.97	85.76 ± 0.75
HookNet	87.14 ± 0.49	90.23 ± 0.44	88.69 ± 0.48	86.47 ± 0.52
SlideFiLM	**88.73 ± 0.48**	**90.81 ± 0.72**	**89.77 ± 0.43**	**91.78 ± 0.47**

Cross-cohort segmentation performance on MARA. Dice similarity coefficients (%) are reported. Patch-level and WSI-level Dice are computed using the same evaluation protocol as in Table 1.

On the KPIs dataset, SlideFiLM achieved the best average Boundary F1 score of 67.08 and the lowest average HD95 of 268.73 pixels among all compared methods (Supplementary Table 1). These results indicate improved boundary agreement and reduced spatial discrepancy at the whole-slide level. SlideFiLM also achieved the highest average object-level F1 of 84.22, suggesting that the proposed slide-conditioned decoder better preserves separated glomerular structures and reduces object-level fragmentation compared with baseline models. These improvements are consistent across the four KPIs cohorts, supporting the claim that SlideFiLM improves not only overlap-based accuracy but also structural completeness.

On the MARA dataset, SlideFiLM again achieved the best structure-aware performance, with the highest average Boundary F1 of 62.16 and the lowest average HD95 of 179.71 pixels (Supplementary Table 2). Because MARA masks typically contain only a small number of large contiguous plaque regions rather than many separated objects, connected-component-based object-level metrics are less informative and can be unstable for this dataset. Therefore, Boundary F1 and HD95 were used as the primary structure-aware metrics for MARA. The results show that SlideFiLM produces more accurate and spatially coherent plaque boundaries than the baseline methods.

Together, these quantitative structure-aware evaluations support the visual findings in Fig. 2. SlideFiLM improves WSI-level segmentation not only in terms of Dice overlap, but also in boundary accuracy, structural consistency, and object-level preservation where component-based evaluation is appropriate.

SlideFiLM enables label-efficient learning under low-data regimes

A major limitation in medical image segmentation is the scarcity of expert pixel-level annotations, particularly for whole-slide images, where dense labeling is manual, time-consuming, and difficult to scale. To quantify label efficiency, we varied the fraction of labeled training slides on the KPIs dataset and the MARA dataset (20%, 40%, 80%, and 100%). We report SlideFiLM together with HookNet, the strongest fully supervised baseline in our benchmark. Each setting was repeated with three random seeds, and results are reported as mean ± standard deviation.

As shown in Fig. 3, SlideFiLM maintained consistently strong cohort-wise performance across training fractions, while HookNet degraded much more sharply in the low-label regime. On KPIs, SlideFiLM achieved 89.83 ± 0.95 patch-level Dice and 87.19 ± 0.81 WSI-level Dice at 20% labeled slides, increasing to 92.96 ± 0.73 and 90.03 ± 0.71 at 100% labels (Supplementary Table 3). In contrast, HookNet dropped to 79.14 ± 3.08 patch-level Dice and 75.36 ± 2.77 WSI-level Dice at 20% supervision, although it recovered to 91.05 ± 0.65 patch-level Dice at full supervision. This pattern was also evident at the cohort level, where HookNet showed larger performance drops and visibly wider error bars, especially in DN, 5/6Nx, and NEP25, whereas SlideFiLM remained comparatively stable.

A similar trend was observed on the MARA dataset. SlideFiLM achieved 86.18 ± 0.68 patch-level Dice and 86.48 ± 0.91 WSI-level Dice at 20% labeled slides, improving to 89.77 ± 0.43 and 91.78 ± 0.47 at full supervision (Supplementary Table 4). HookNet again showed a larger drop under limited supervision, reaching 82.53 ± 2.81 patch-level Dice and 79.52 ± 2.06 WSI-level Dice at 20%, and remained below SlideFiLM even at 100% labels. Overall, these results show that SlideFiLM is more label-efficient and more stable when annotation budgets are limited, while HookNet is more sensitive to reduced supervision and exhibits greater variance in low-data settings.

Taken together, these results show that SlideFiLM attains state-of-the-art performance across multiple WSI datasets, combining high pixel-level accuracy with structural completeness. The consistent gains across both datasets and training fractions further support the effectiveness of slide-level conditioning for label-efficient WSI segmentation under limited supervision.

Ablation Study: Effect of Freezing Strategy

To determine how much adaptation is necessary under limited supervision, we ablated the encoder freezing strategy while keeping all other training conditions fixed. Specifically, we compared: (1) All-Frozen (AF), which preserves both the slide-level encoder and the tile backbone and trains only the decoder; (2) Partial-Unfreeze (PU-TileLast), which keeps the slide encoder frozen but unfreezes only the final stage of the tile backbone; and (3) All-Unfrozen (AU), which enables full end-to-end fine-tuning.

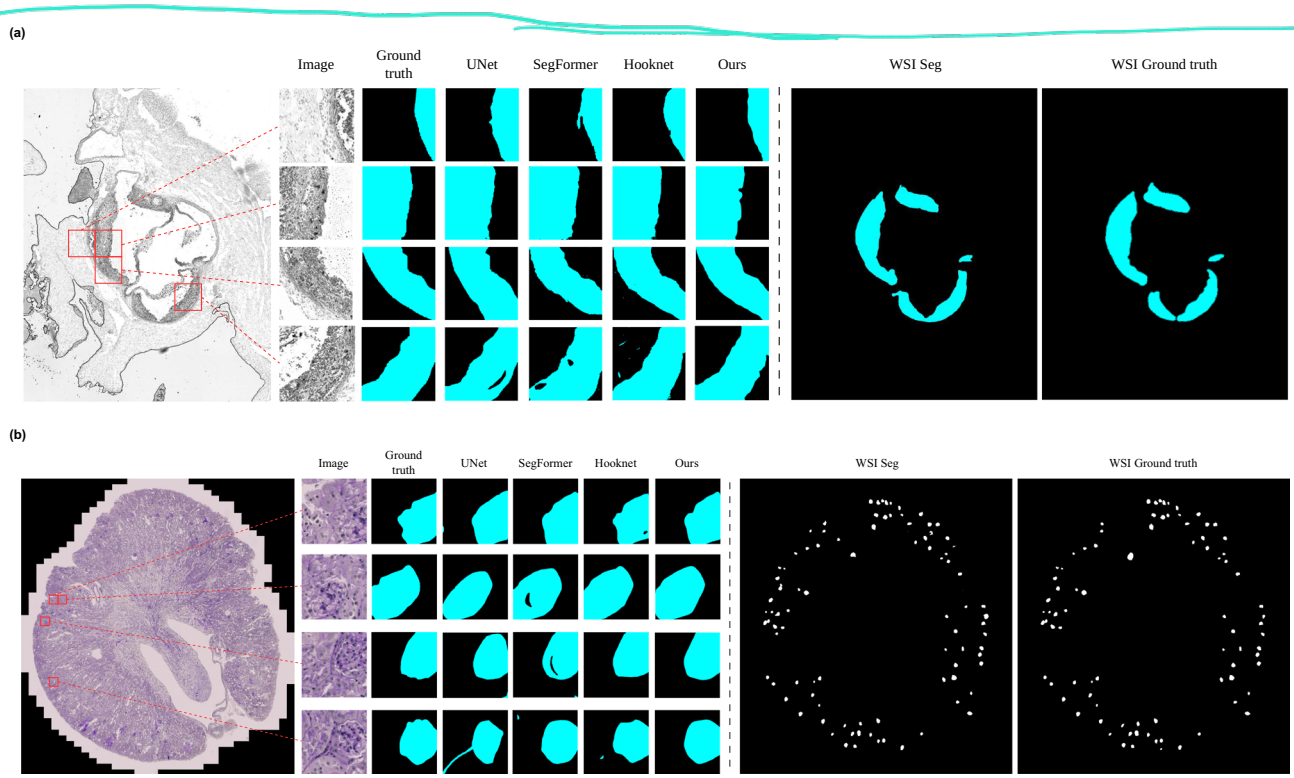


Fig. 2 | Qualitative comparison of WSI segmentation results on the MARA and KPIs datasets. a MARA WSI. **b** KPIs WSI. For each WSI, four representative zoom-in regions are selected to highlight fine-grained structures. From left to right: input image patches, ground-truth annotations, and predictions from U-Net, SegFormer, HookNet, and SlideFiLM. The rightmost panels show full-slide segmentation results aggregated from tile-wise predictions. Compared with the baseline models, SlideFiLM yields more accurate boundaries, better structural completeness, and fewer spurious fragments across both datasets.

Across KPIs and MARA, AF provided stable and consistently high performance in the low-data regime, confirming that pretrained global histomorphological priors are directly transferable when annotation is scarce. Introducing limited adaptability in PU-TileLast yielded small, consistent gains once more labeled WSIs were available. This suggests that, as supervision increases, the model mainly needs to adjust fine-grained local texture cues, whereas the global slide-level structure is already well captured and requires little additional tuning. By contrast, AU exhibited substantially higher variance and degraded performance when training data were insufficient, indicating that full end-to-end updates can disrupt pretrained hierarchical organization and lead to catastrophic forgetting rather than beneficial adaptation.

While the relative behavior of the three strategies varies with supervision, one observation was consistent in the low-data regime: AF delivered the most reliable and competitive performance when labels were limited, whereas partial unfreezing (PU-TileLast) became a reasonable option when additional labeled slides were available. Because this pattern held across training fractions, we report the representative full-data comparison in Supplementary Table 5 and Table 6. These results support AF as the default optimization strategy for low-data regime WSI segmentation, with partial unfreezing as a controlled mechanism for targeted adaptation when additional labeled data are available.

Ablation study on the contribution of architectural components

In our model, improvements can originate from a strong pretrained encoder or from the segmentation architecture that converts encoder features into dense predictions. This raises the concern that the improvements of SlideFiLM might be largely attributed to the pretrained backbone rather than to the proposed architectural design. To investigate this, we conducted an ablation study on two datasets, keeping the encoder initialization, training protocol, and evaluation pipeline fixed across all settings. We tested the impact of the key architectural components by comparing four variants: an encoder-decoder baseline, a variant without FiLM, a variant without skip connections, and the full SlideFiLM model.

As shown in Supplementary Table 7 and Table 8, FiLM conditioning accounts for the majority of the improvement. Removing FiLM largely collapses the model back toward the baseline performance, indicating that the proposed gains are not explained by encoder pretraining alone and require conditional feature modulation in the segmentation head. In contrast, removing skip connections yields a smaller but consistent degradation relative to the full model, suggesting that multi scale feature fusion provides additional refinement rather than being the primary driver of accuracy. Notably, the full model performs best under both patch level and whole slide evaluation, supporting the finding that FiLM improves the quality of local predictions and also enhances global consistency when aggregating predictions at slide scale.

These observations indicate that FiLM benefits arise from its ability to adaptively modulate decoder features in a class dependent manner, improving separation under appearance heterogeneity and reducing ambiguity in challenging structures. Skip connections further contribute by injecting spatially detailed features, which helps refine boundaries and recover fine structures, but their effect is secondary compared with FiLM in this setting. In summary, the ablation results demonstrate that the advantage of SlideFiLM is driven primarily by the proposed architectural design, with FiLM providing the core gain and skip connections offering complementary improvements.

Ablation study on slide-level context encoder

Since SlideFiLM relies on slide-level contextual representations to modulate tile-level decoding, we further examined whether the choice of foundation model affects segmentation performance. In addition to the original Prov-GigaPath slide representation, we evaluated two recent ViT-based pathology foundation models, Virchow[25] and UNI2[26], as alternative context encoders. Because Virchow and UNI2 primarily provide tile-level

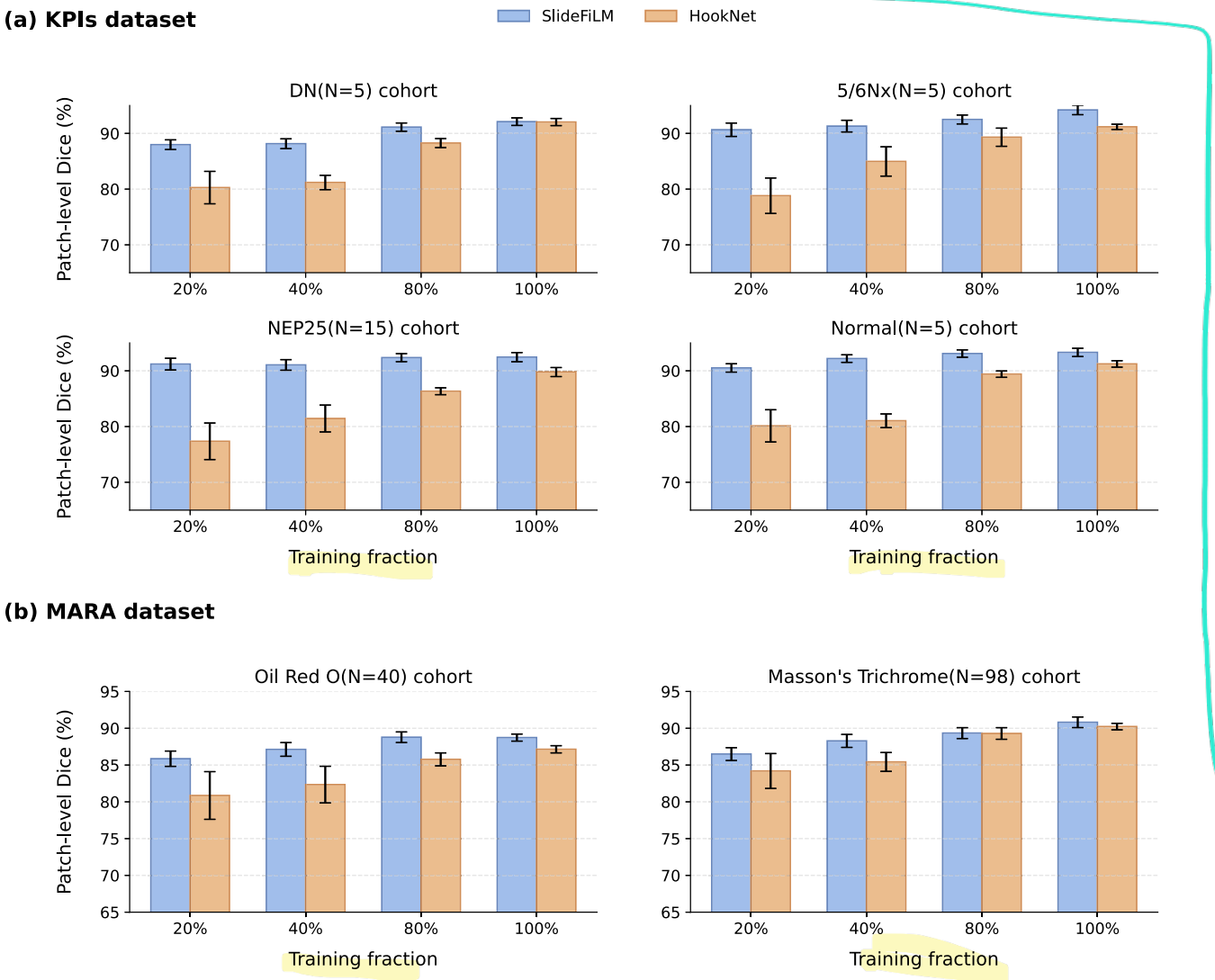


Fig. 3 | Label-efficiency comparison of SlideFiLM and HookNet across different training fractions. a Patch-level Dice on the KPIs dataset. **b** Patch-level Dice on the MARA dataset. Bars indicate the mean Dice across three runs, and error bars denote the standard deviation.

representations in our implementation, we extracted tile embeddings from all non-overlapping tiles of each WSI and applied mean pooling to obtain a single slide-level context vector. This pooled representation was then used as the FiLM conditioning input, while the segmentation decoder, FiLM modulation mechanism, training settings, data splits, post-processing pipeline, and evaluation protocol were kept unchanged. Therefore, this ablation isolates the effect of the slide-level context representation.

As shown in Supplementary Table 9, on the KPIs dataset, the Prov-GigaPath-based variant achieved the strongest overall segmentation performance, with the highest average WSI-level Dice of 90.74 and the highest patch-level Dice of 93.74. It also achieved the best average Boundary F1 score of 67.08, indicating improved boundary agreement and structural consistency. Virchow-Mean produced a slightly lower average HD95 than Prov-GigaPath, and UNI2 achieved the highest object-level F1 score. However, Prov-GigaPath provided the best overall balance between region overlap, patch-level accuracy, and boundary-aware performance across the four KPIs cohorts.

The advantage of Prov-GigaPath was more pronounced on the MARA dataset (Supplementary Table 10). Compared with Virchow-Mean and UNI2, the Prov-GigaPath variant achieved the highest average WSI-level Dice, patch-level Dice, Boundary F1, and the lowest HD95. Specifically, Prov-GigaPath improved the average WSI-level Dice from 82.95 with Virchow-Mean and 85.89 with UNI2 to 91.14, while reducing the average HD95 from 369.30 and 290.57 to 179.71. These results suggest that the explicit whole-slide representation learned by Prov-GigaPath provides more informative global tissue context than simple mean pooling of tile-level foundation embeddings.

Overall, this ablation supports the use of Prov-GigaPath as the default slide-level context encoder in SlideFiLM. The results indicate that the performance gain is not merely due to using any pretrained pathology ViT representation, but is strengthened by a slide-level representation that explicitly models long-range dependencies across the WSI. This finding is consistent with the motivation of SlideFiLM: dense tile-level predictions benefit from global slide context when the context representation captures whole-slide tissue organization rather than only averaged local tile semantics.

Ablation study on the FiLM boundary parameter

The bounded FiLM formulation introduces a boundary parameter λ to control the modulation range of the slide-conditioned scale and shift parameters. Since this parameter determines the strength of slide-level conditioning, we conducted a sensitivity analysis to evaluate whether the performance of SlideFiLM depends strongly on a specific λ value. We compared bounded FiLM with $\lambda \in \{0.10, 0.25, 0.50, 1.00\}$, together with an unbounded FiLM variant in which the tanh-based constraint on the FiLM scale and shift parameters was removed. All variants were trained and

evaluated using the same data splits, training protocol, post-processing pipeline, and evaluation metrics.

As shown in Supplementary Table 11, on the KPIs dataset, all bounded FiLM variants remained numerically stable throughout training, with no NaN losses observed. The best patch-level Dice was obtained with $\lambda = 1.00$, while the best WSI-level Dice was obtained with $\lambda = 0.25$. Importantly, $\lambda = 0.50$ achieved near-best performance on both metrics, with 92.96

A similar trend was observed on the MARA dataset (Supplementary Table 12). All bounded variants completed training without NaN losses, whereas the unbounded variant produced NaN losses at Epoch 24. For MARA, $\lambda = 0.25$ achieved the highest patch-level Dice, while $\lambda = 0.50$ achieved the highest WSI-level Dice. The difference between $\lambda = 0.25$ and $\lambda = 0.50$ at the patch level was marginal, 89.81% versus 89.77%, while $\lambda = 0.50$ provided the strongest WSI-level performance. This result suggests that moderate bounded conditioning provides a favorable balance between local prediction accuracy and whole-slide consistency.

Overall, these results support the use of the bounded FiLM formulation for stable mixed-precision optimization. The unbounded variant was prone to activation divergence and NaN losses, whereas all bounded settings were stable. We therefore used $\lambda = 0.50$ as the default value in all main experiments because it provides a balanced modulation range, avoids the instability of unbounded FiLM, and achieves best or near-best performance across both patch-level and WSI-level evaluations.

Discussion

This study presents SlideFiLM as a slide-conditioned framework for data-efficient whole-slide image segmentation. The main contribution is the formulation of dense WSI segmentation as a slide-to-tile conditioning problem. SlideFiLM decouples global context from the local image branch by deriving a compact whole-slide representation from a pretrained WSI foundation model and using it as a non-spatial conditioning signal for tile-level decoding. This design allows every tile from the same slide to be decoded under a shared representation of slide-wide tissue organization, rather than relying only on the local crop or a parallel contextual image crop.

The proposed bounded FiLM formulation further distinguishes SlideFiLM from generic conditional modulation. Standard FiLM can freely scale and shift features, which may destabilize optimization when the conditioning signal is shared across many tiles and when only limited dense annotations are available. SlideFiLM uses an identity-centered bounded modulation, restricting the magnitude of slide-conditioned scale and shift parameters. This design allows the slide embedding to adjust pretrained tile features without overwriting them, thereby preserving the stability of the foundation representation while still enabling slide-aware adaptation. The sensitivity analysis showed that bounded FiLM variants remained stable across datasets, whereas the unbounded variant was prone to numerical instability. Thus, the conditioning module is not only a direct application of FiLM, but a constrained slide-conditioned adaptation mechanism designed for low-data WSI segmentation.

The proposed segmentation framework showed strong performance across all training fractions, including the fully supervised setting, while its advantage was particularly evident under limited annotation. In these low-data regimes, SlideFiLM remained comparatively stable. This suggests that slide-level contextual priors serve as an implicit regularizer, narrowing the space of feasible tile-level predictions and reducing reliance on large annotated datasets. The ablation studies support this interpretation, as FiLM conditioning accounted for the main performance gain, skip connections provided additional refinement of boundaries and fine structures, and freezing the pretrained encoders was the most reliable optimization strategy in low-data settings. In practice, this property directly improves the feasibility of deploying segmentation models in real clinical or research settings, where annotations are scarce and expensive.

SlideFiLM demonstrated consistent gains across both public and newly curated data. Importantly, validation on the newly developed expert-annotated MARA dataset strengthens the clinical relevance of the study by showing that the model is effective not only on public benchmark data but also on a real-world cohort with specialist annotations. This additional dataset expands the empirical basis of the paper from methodological benchmarking toward real-world validation, which is especially valuable in computational pathology where dense expert labels remain scarce.

There are several limitations to this work. First, our experiments focus on the quantification of renal pathology and atherosclerosis, and a broader validation across tissue types, staining protocols, and acquisition domains will be required to evaluate generality at scale. Second, the current implementation produces deterministic segmentation maps. Incorporating uncertainty estimation or calibrated confidence may facilitate clinical review and highlight ambiguous regions[27,28]. Finally, although the FiLM mechanism improves consistency across tiles, future work may explore explicit multi-resolution fusion to better integrate cellular morphology with slide-level context. Despite these limitations, our results underscore the importance of integrating global tissue context into dense whole-slide segmentation. This strategy may support more robust and data-efficient computational pathology models for gigapixel histology analysis.

In conclusion, SlideFiLM provides a flexible foundation for context conditioned learning in whole-slide segmentation. Potential directions include incorporating multi-resolution fusion, refining slide embeddings to emphasize disease-salient features, and developing interactive annotation workflows where slide-level context guides both automated predictions and human review. More broadly, our findings reinforce the importance of integrating global tissue context into histopathology models and suggest that slide-conditioned decoding may play a key role in enabling accurate segmentation that remains effective with limited training data at gigapixel scale.

Methods

Datasets

KPIs The Kidney Pathology Image Segmentation (KPIs) cohort includes 60 PAS-stained WSIs obtained from 20 rodents, covering four mouse models representing distinct chronic kidney disease (CKD) conditions: 5/6Nx, DN, NEP25, and Normal. All slides were scanned at 40 × magnification using a Leica SCN400 system. This dataset provides substantial morphological diversity across CKD stages and serves as a challenging benchmark for segmentation performance.

MARA The Ouimet Lab Mouse Aortic Root Atherosclerosis dataset (MARA) is a newly annotated real-world histology dataset generated at the Ottawa Heart Institute. It comprises digitized images of murine aortic root sections from two experimental cohorts. All animal procedures were approved by the University of Ottawa Animal Care and Use Committee (protocol HIe-3942) and performed in accordance with relevant guidelines and regulations.

Cohort 1 included 142 images from 16 *ApoE*$^{-/-}$ mice (Jackson Laboratory, B6.129P2-*Apoe1Unc*/J; strain 002052; 10 males, 6 females). Mice were housed in a temperature- and light-controlled environment and maintained on a standard laboratory chow diet (Envigo 2019) until study initiation. Starting at 8 weeks old, animals were fed a Western diet (Envigo TD.88137, 0.2% cholesterol) for 8 weeks and randomized to anesthesia only, surgery, or surgery with rApoA-I treatment (single intraperitoneal injection, 40 mg kg^{-1} at emergence) based on weights and plasma cholesterol measurements (Infinity™ Cholesterol Liquid Stable Reagent, Thermo Scientific, TR13421). Surgery consisted of abdominal laparotomy under 2 − 3% isoflurane anesthesia as previously described[29]. Fifteen days after surgery, mice were euthanized under 3 − 4% isoflurane anesthesia by cardiac puncture followed by PBS perfusion, and tissues were collected. Blinding was maintained whenever possible during the experiment, and throughout sample processing and analysis. Cohort 2 included 56 images from 7 mice aged 10 weeks (WT or Osbpl6$^{-/-}$) injected with 1×10^{11} AAV-PCSK9 gain-of-function particles and fed a Western diet for 12 weeks to induce atherosclerosis before tissue collection at the study endpoint.

Aortic roots were harvested, embedded in OCT, flash-frozen, and stored at − 80 °C. Aortic sinuses were serially cryosectioned at 10 μm thickness at 100 μm intervals and stored at − 20 °C. One slide per animal

was stained, using Masson's trichrome for cohort 1 and Oil Red O for cohort 2. Slides containing 8–14 sections were digitized using a Leica Aperio 8 scanner with an HC Plan-Apochromat 20 × /0.8 objective, yielding one image per section.

Manual image analysis and annotation were performed in Fiji[30]. Plaques were defined as lesions located within the valve leaflets of the aortic sinus, whereas regions caudal to the valve, outside leaflet boundaries, and non-atherosclerotic vessel wall were excluded. Plaques were quantified from first appearance to disappearance, up to 1,200 μm beyond valve onset at 100 μm intervals. For model development, each image was stored as a raw TIF file with a corresponding manually annotated ROI mask in TIF format.

Preprocessing and data splitting

During data preprocessing, all WSIs were tiled into fixed-size patches for model development. For the KPIs dataset, the authors' predefined training, validation, and test splits were used. For the MARA dataset, splitting was performed at the animal level using the unique mouse identifiers. All histological images derived from the same mouse were assigned to the same subset, ensuring that no images from the same animal appeared across the training, validation, and test sets. Approximately 70% of the mice were assigned to the training set, 15% to the validation set, and the remaining 15% to the test set. This animal-level splitting strategy was used to prevent subject-level data leakage. For the experiments under low-data regimes, 20%, 40%, 80%, and 100% of the annotated WSIs were sampled from the training or development split, while the corresponding fixed test split was retained for final performance assessment. To ensure consistency with the resolution used during Prov-GigaPath pretraining (0.5 μm/pixel), all whole-slide images were first downsampled to the corresponding pyramid level before tiling. During training, each slide was partitioned into non-overlapping 256 × 256 px RGB tiles to avoid redundant sampling and ensure that each spatial region contributed equally to optimization. For inference, we adopted a 50% overlap tiling strategy to mitigate border artifacts and improve spatial coherence near tile boundaries. Overlapping tiles were independently processed by the model, and the outputs were aggregated using an averaging scheme in the overlapping regions before reconstructing the full-resolution prediction map. This approach provides smoother slide-level predictions, particularly for elongated structures and fine-grained tissue interfaces.

Network architecture

The proposed SlideFiLM framework builds upon the Prov-GigaPath foundation model, extending it from representation learning to pixel-level whole-slide segmentation. Specifically, each WSI is first partitioned into non-overlapping 256 × 256 tiles, which are individually processed by a ViT pretrained using the DINOv2 self-supervised framework[31–33]. This tile-level encoder converts each tile into both patch-level embeddings that retain local spatial information and a global image token ([CLS]) summarizing tile semantics. All [CLS] tokens are aggregated by the Prov-GigaPath slide-level encoder, implemented with dilated attention (LongNet), to model ultra-long contextual dependencies across the entire slide[34]. The resulting slide-level embedding encodes holistic tissue context and relationships between tiles across gigapixel scales. This embedding is then fed into a FiLM parameter generator, which outputs modulation coefficients (γ, β) for each decoder level.

The decoder follows a multi-stage FiLM-modulated upsampling process. At each decoder block, the current feature map is first modulated by the corresponding FiLM parameters. Each block then performs upsampling, convolution, and group normalization followed by GELU activation and refinement. To preserve spatial fidelity, SlideFiLM integrates skip connections from selected intermediate ViT layers, specifically layers 5, 11, 23, and 39, corresponding respectively to early, shallow-mid, deep-mid, and late semantic stages of the encoder. Before being fused into the decoder, these skip features are passed through an attention layer, which adaptively reweights the intermediate ViT representations to highlight task-relevant spatial and semantic cues while suppressing less informative responses[35]. The attended multi-scale skip features are then spatially aligned and concatenated with the upsampled decoder features at matching resolution levels, enabling progressive fusion of fine-grained textures and high-level contextual information throughout reconstruction. The final segmentation head, equipped with an additional FiLM modulation, produces pixel-aligned probability maps representing the local tissue mask.

This design allows the model to propagate slide-level semantics through every decoding stage, aligning global histopathological context with local morphology. By coupling Prov-GigaPath's foundation-scale contextual understanding with FiLM-based modulation, SlideFiLM enables slide-level context to guide tile-level decoding while preserving local morphological detail.

Training protocol and experimental setting

To balance morphological diversity and training stability, an inverse cosine-based sampling strategy was employed to control the proportion of background tiles throughout training. At the beginning of training, the model was exposed to a higher proportion of background or low-tissue tiles to stabilize early optimization and prevent overfitting to high-frequency textures. As training progressed, the sampling probability of background tiles gradually decreased following an inverse cosine annealing schedule:

$$r_t = r_{\max} - \frac{1}{2}(r_{\max} - r_{\min})\left[1 - \cos\left(\frac{\pi t}{T}\right)\right], \quad (1)$$

where r_t denotes the ratio of background tiles sampled at epoch t, $r_{\max}$ and $r_{\min}$ are the initial and final sampling ratios, and T is the total number of training epochs. This setup allowed us to systematically evaluate the model's performance under varying amounts of annotated data, while maintaining independent validation sets for unbiased performance assessment. The progressive reduction in background sampling further encouraged the model to focus on diagnostically relevant tissue regions, thereby enhancing feature discriminability and improving robustness across heterogeneous slides and staining conditions.

The model was built upon the pretrained Prov-GigaPath foundation. To leverage its rich representation capability while maintaining computational efficiency, the encoder weights were kept frozen during training, and only the decoder parameters were updated. The decoder, composed of upsampling and feature-fusion blocks, was trained to adapt the fixed slide-level embeddings into pixel-aligned segmentation outputs. This design enables efficient adaptation of foundation-scale knowledge to the target segmentation task while preserving the pretrained representation.

The decoder was optimized using the AdamW[36] optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with an initial learning rate of 1×10^{-5} and a weight decay of 1×10^{-4}. Mixed-precision training was performed using gradient scaling to improve numerical stability[37]. The training objective was defined as the sum of the binary focal loss ($\alpha = 0.25$, $\gamma = 2.0$)[38] and the soft Dice loss[39]. For binary segmentation, the decoder outputs a single-channel probability map with a sigmoid activation, which is binarized at a threshold of 0.5 to obtain the final foreground mask. The slide batch size was set to 1 and the tile batch size was set to 32, corresponding to one whole-slide image and its associated tile set per iteration. To enhance generalization across tissue types and staining variations, several geometric and photometric augmentations were applied. These augmentations simulate real-world variability in tissue orientation and staining conditions, encouraging the model to learn stain- and rotation-invariant features. The model was trained on a single NVIDIA A100 (40 GB) GPU. Because our framework leverages the pretrained Prov-GigaPath tile features and slide-level contextual embeddings, the segmentation decoder operates on high-level semantic representations rather than raw image patches. This substantially simplifies the optimization landscape and leads to rapid convergence. Consequently, we adopted a short training schedule of 50 epochs, selecting the checkpoint with the highest validation Dice score.

During inference, each whole-slide image was partitioned into partially overlapping tiles to reduce boundary artifacts introduced by tile-wise prediction. Tile outputs were stored as grayscale probability maps and stitched

into a full-resolution canvas according to their recorded spatial offsets. In overlapping regions, probabilities were aggregated by averaging across all contributing tiles. The stitched probability map was further smoothed with a Gaussian filter in probability space using $\sigma = 8$, after which a fixed threshold of 0.5 was applied to obtain a binary mask.

To further improve the robustness of the final segmentation, a lightweight post-processing pipeline composed of standard morphological operations was applied. Specifically, connected component analysis was performed on the stitched binary mask to remove small isolated predictions, retaining only components above a minimum area threshold of 20,000 pixels. Subsequently, morphological hole filling was applied with an area threshold of 100,000 pixels to each retained component to suppress spurious internal voids caused by local prediction uncertainty and enforce spatial coherence within contiguous pathological regions. Finally, the mask boundary was regularized using morphological opening followed by morphological closing with a disk-shaped structuring element of radius 2 pixels, which reduces small spurs and notch-like irregularities while preserving the overall lesion extent. This post-processing strategy preserves clinically meaningful continuous structures while effectively eliminating scattered false positives and internal fragmentation, leading to improved spatial consistency and more stable downstream quantitative evaluation.

To ensure a fair comparison with baseline models, all models were trained under the same data splits, preprocessing pipeline, WSI reconstruction procedure, post-processing pipeline, and evaluation protocol. Shared training settings, including data splits, input resolution, tiling strategy, loss function, optimizer, and evaluation metrics, were kept consistent where applicable, while architecture-specific hyperparameters and implementation details for each baseline were adopted from the original publications. For WSI-level evaluation, tile-wise predictions from Slide-FiLM and all baseline models were mapped back to the slide coordinate space using the same stitching strategy. The same probability aggregation, Gaussian smoothing, fixed thresholding, connected-component filtering, hole filling, and morphological steps were then applied before metric computation. The post-processing hyperparameters were fixed based on the validation set and were not tuned separately for individual models or test slides.

Slide-conditioned FiLM module

In our setting, the slide embedding is shared by all tiles from the same WSI and directly controls decoder feature modulation. If the FiLM scale and shift are left unconstrained, the conditioning signal may excessively amplify, suppress, or shift pretrained tile features, causing unstable activations and inconsistent slide-level reconstruction, especially under scarce annotations and mixed-precision training. We therefore use an identity-centered bounded FiLM formulation, where slide-level context can adjust decoder features within a controlled range rather than overwrite the pretrained representation. In the classical formulation, given features $\mathbf{F} \in \mathbb{R}^{B\times C\times H\times W}$ and conditioner $\mathbf{c}$, FiLM applies

$$\mathbf{Y}_{\text{orig}} = \boldsymbol{\Gamma}(\mathbf{c}) \odot \mathbf{F} + \boldsymbol{\beta}(\mathbf{c}), \qquad \boldsymbol{\Gamma}, \boldsymbol{\beta} \in \mathbb{R}^{B\times C}, \tag{2}$$

with unconstrained scale and shift[21]. In our setting, such freedom can suppress features ($\boldsymbol{\Gamma} \approx \mathbf{0}$) or induce unstable gains ($\boldsymbol{\Gamma} \gg 1$) and offers no control over inter-tile consistency.

We therefore adopt an identity-centered, bounded variant that conditions on the slide embedding $\mathbf{s}$:

$$\mathbf{Y} = (1 + \boldsymbol{\gamma}(\mathbf{s})) \odot \mathbf{F} + \boldsymbol{\beta}(\mathbf{s}), \qquad \|\boldsymbol{\gamma}\|_\infty, \|\boldsymbol{\beta}\|_\infty \le \lambda. \tag{3}$$

We implement the bound via a tanh gate scaled by λ and set $\lambda = 0.5$ in all experiments, which restricts the per-channel gain to $[1-\lambda,\ 1+\lambda] = [0.5,\ 1.5]$ and the bias to $[-0.5,\ 0.5]$.

For a per-location vector $\mathbf{f} \in \mathbb{R}^C$, write $\mathbf{y} = \mathbf{A}\mathbf{f} + \boldsymbol{\beta}$ with $\mathbf{A} = \text{diag}(1 + \boldsymbol{\gamma})$. The bounds $1-\lambda \le 1+\gamma_i \le 1+\lambda$ imply

$$\sigma_{\min}(\mathbf{A}) \ge 1-\lambda, \quad \|\mathbf{A}\|_2 \le 1+\lambda, \quad \kappa(\mathbf{A}) \le \frac{1+\lambda}{1-\lambda}, \tag{4}$$

so the feature Jacobian is well conditioned across decoder stages. Deviation from the pretrained representation is explicitly limited,

$$\|\mathbf{y} - \mathbf{f}\|_2 = \|\boldsymbol{\gamma} \odot \mathbf{f} + \boldsymbol{\beta}\|_2 \le \lambda \|\mathbf{f}\|_2 + \lambda\sqrt{C}, \tag{5}$$

which keeps optimization near the Prov-GigaPath feature space while permitting slide-aware reweighting of class-informative channels. Practically, $\boldsymbol{\gamma}(\mathbf{s})$ and $\boldsymbol{\beta}(\mathbf{s})$ are shared across tiles of a slide and broadcast spatially, reducing logit discontinuities at tile borders and improving stitching. The identity start and bounded magnitudes provide explicit safeguards that markedly lower NaNs induced by overflow under mixed precision.

Evaluation metrics

Segmentation performance was evaluated using Dice, Boundary F1, HD95, and object-level F1. The Dice coefficient was used as the primary overlap metric and was reported at both the patch and whole-slide image (WSI) levels.

The Boundary F1 score measured contour agreement using a tolerance of 10 pixels, while the HD95 quantified boundary displacement. Higher Dice and Boundary F1 values and lower HD95 values indicate better segmentation performance.

The object-level F1 score evaluated the detection of individual structures. Predicted and ground-truth connected components were matched one-to-one using an IoU threshold of 0.5. This metric was reported only for the KPIs dataset because the MARA annotations contained few large connected lesion regions.

Loss function

To handle the severe foreground-background imbalance inherent in WSI segmentation, we adopt the binary focal loss. The focal loss dynamically reduces the contribution of easy negatives while emphasizing hard examples, a mechanism designed specifically for highly imbalanced dense-prediction tasks. To further improve overlap-based optimization, we incorporate the soft Dice loss[39], which has been consistently demonstrated to be robust and effective for medical image segmentation in prior work[40]. The final objective is defined as the unweighted sum of focal loss and Dice loss. Let S and G denote the predicted segmentation and ground truth, respectively. For pixel i, $s_i \in [0, 1]$ and $g_i \in \{0, 1\}$ represent the predicted probability and ground-truth label. N denotes the total number of pixels in the image.

The binary focal loss is defined as

$$L_{\text{focal}} = -\frac{1}{N}\sum_{i=1}^{N}\left[\alpha\, g_i (1-s_i)^{\gamma}\log(s_i) + (1-\alpha)(1-g_i)\, s_i^{\gamma}\log(1-s_i)\right], \tag{6}$$

where $\alpha \in [0, 1]$ is the balancing factor between positive and negative classes, and $\gamma \ge 0$ controls the degree of focusing on hard samples. In our experiments, we set $\alpha = 0.25$ and $\gamma = 2.0$.

The binary Dice loss is defined as

$$L_{\text{Dice}} = 1 - \frac{2\sum_{i=1}^{N} s_i g_i + \varepsilon}{\sum_{i=1}^{N} g_i^2 + \sum_{i=1}^{N} s_i^2 + \varepsilon}, \tag{7}$$

where ε is a small constant to avoid division by zero.

The final loss used during training is the sum of the two terms:

$$L = L_{\text{focal}} + L_{\text{Dice}}. \quad (8)$$

Data Availability

The MARA dataset generated during the current study is not publicly available due to institutional data governance restrictions but is available from the corresponding author on reasonable request.

Code availability

The code used to train, validate and evaluate the model in this study is publicly available at https://github.com/Lunchi-Kwok/SlideFilm.

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Author contributions
L.G., D.B., M.I.O. and C.S. conceived the study and contributed to the study design. L.G. and C.S. developed the methodology. D.B. and M.I.O. contributed clinical and histopathological expertise and supported the biological and domain-specific interpretation of the results. L.G. implemented the model, performed the experiments and formal analyses, and prepared the figures. L.G. and C.S. wrote the manuscript with input from all authors. C.S. supervised the study. All authors reviewed and approved the final manuscript.

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Correspondence and requests for materials should be addressed to Christopher Sun.

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